A novel Convolutional Tsetlin Architecture for classification of Satellite images under limited supervision

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Abstract

Tsetlin Machines (TMs) provide propositional rule-based learning that is natively interpretable, yet their application to high-dimensional remote sensing imagery under limited supervision remains largely unexplored. This paper introduces, to the best of our knowledge, the first unified CNN–Tsetlin Machine (CNN–TM) pipeline for multi-modal satellite image classification in low-data conditions.Multi-scale convolutional front-end allows the extraction of detailed spatial features, which are represented using a concise Boolean representation technique called structure binary encoding. It captures relative intensity, texture variation, and scattering capability. The TM extracts propositions using this representation technique, enabling classification along with reasoning behind prediction within the same framework without any approximation afterward. This model's performance is tested on four benchmark data sets comprising MSTAR-10 (SAR), EuroSAT (multi-spectral), SAMPLE-10 (simulated SAR), and Indian Pines (hyperspectral). The experiment was performed under the condition of having only 121 labeled samples for each class. Under this constraint, this framework achieved an incredible 98.39% accuracy on MSTAR-10, outperforming eleven cutting-edge deep learning frameworks.

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Abbreviations

ATR	Automatic Target Recognition
CNN	Convolutional Neural Network
TM	Tsetlin Machine
CTM	Convolutional Tsetlin Machine
CNN–TM	Hybrid CNN–Tsetlin Machine framework proposed in this work
SAR	Synthetic Aperture Radar
HSI	Hyperspectral Imaging
LULC	Land Use and Land Cover classification
RGB	Red–Green–Blue image channels
BN	Batch Normalisation layer
ReLU	Rectified Linear Unit activation function
CV	Coefficient of Variation (texture descriptor)
Optuna	Automated hyperparameter optimisation framework
GradCAM	Gradient-weighted Class Activation Mapping (post-hoc explanation method)
LIME	Local Interpretable Model-agnostic Explanations (post-hoc explanation method)
AVIRIS	Airborne Visible/Infrared Imaging Spectrometer (hyperspectral sensor)
SVM	Support Vector Machine classifier
RBF	Radial Basis Function kernel
ViT	Vision Transformer architecture
MSTAR	Moving and Stationary Target Acquisition and Recognition dataset

Symbols

$Z \in \mathbb{R}^{C \times H \times W}$	CNN feature map output C
Number of feature channels H, W	Spatial height and width of feature map P_{ij}^{c}
Patch at position (i, j) in channel c μ_c	Mean of channel c σ_c
Standard deviation of channel c μ_{ij}	Mean of patch P_{ij}^{c} σ_{ij}
Standard deviation of patch P_{ij}^{c} m_{ij}	Maximum activation in patch P_{ij}^{c} μ_{rel}
Relative mean (patch-to-channel ratio) σ_{rel}	Relative standard deviation cv
Coefficient of variation of patch $b^{(n)} \in 0, 1^{d}$	Binary feature vector of sample n d
Dimension of binary feature vector (d = 36,864) g_{ψ}	CNN feature extraction function ϕ
Structured binary encoding function f_{θ}	Tsetlin Machine classifier ψ
Parameters of CNN feature extractor θ	Parameters of Tsetlin Machine C_{TM}
Number of clauses in Tsetlin Machine T	Voting threshold parameter s
Specificity parameter controlling clause granularity A_n	Set of active (value = 1) binary feature indices for sample n H_n
Spatial attribution heatmap for sample n w_k	Contribution weight of feature k in attribution map ϵ
Small constant for numerical stability p	Pooled feature representation from CNN W_{cls}
Auxiliary classifier weights (used during CNN training only) L	Training loss (cross-entropy with label smoothing) $\hat{y}$
Predicted class label y	Ground truth label

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Chapter 1

Introduction

1.1 Overview

Remote sensing is one of the most used technologies that allows observation and analysis of huge volumes of information from the surface of the Earth. Nowadays, modern satellites produce huge volumes of data using various sensors, including multispectral, hyperspectral, and Synthetic Aperture Radar (SAR). All those sensors contribute to a more detailed understanding of the Earth's surface through its material, texture, and structure characteristics and allow performing such tasks as land use classification, environmental monitoring, agricultural analysis, disaster management, and military purposes.

In the wake of the growing use of satellite imagery, there has been raised an urgent necessity for automation of the interpretation process. Manual analysis of the images was not possible any longer due to the high volume of data involved; therefore, machine learning algorithms have been introduced, especially, CNNs. These networks turned out to be very efficient in feature extraction and achieved unprecedented levels of accuracy.

Though efficient, deep learning models have two main limitations. First, they are black boxes that provide no clue on how the prediction is made. In machine learning applications, transparency becomes an important issue, particularly in applications where the decision made has important implications. Second, deep learning methods require a lot of labelled data during training. Labeling data is expensive and can be hard as well. It may even need expertise to label the data.

In order to address these limitations, it is important to look for a solution that combines the strengths of deep learning and interpretable learning methods.

1.2 Remote Sensing in Visible Spectrum

Remote sensing techniques acquire data based on the reflected electromagnetic waves from the Earth's surface. The visible light spectrum, which includes the red, green and blue bands, is particularly important for identifying different land-cover types. Different materials have distinct reflectance patterns; for instance, vegetation is highly reflective in the green band, water absorbs most of the visible light, and urban materials have variable reflectance patterns based on their composition.

Although spectral information is useful for classification, it can be insufficient on its own. For many satellite images, particularly those with moderate or coarse spatial resolution, one pixel can contain more than one land-cover type. This is known as the mixed-pixel problem and leads to uncertainty and degraded classification performance. So, classification using satellite imagery should not be based solely on spectral information, but rather on spatial information and structure.

1.3 Imaging with Satellite Images

Satellite images offer continuous coverage over large areas, which is ideal for large-scale monitoring. Satellite sensors today collect information across multiple wavelengths, allowing for the extraction of a wide range of information regarding the Earth's surface.

But satellite images come with an inherent trade-off between spatial resolution, temporal resolution and spatial coverage. This often leads to low spatial resolution for many popular data sets, such as multispectral satellite imagery. This results in a loss of fine-scale details like object boundaries, textures and small structures, which are essential for separating visually similar classes.

Limited spatial resolution not only results in lower visual quality, but it also affects feature learning. Training models on such images can make it difficult to learn discriminative features, especially in complex scenes where multiple classes are present in small areas.

1.4 Feature Learning and Representation Issues

The key challenge in satellite image classification is to learn representations. Deep neural networks, such as CNNs, can learn intricate spatial features and hierarchical representations. But they require substantial labelled data for training.

For remote sensing, the cost of data labelling is high, making labelled data more difficult to obtain. With limited supervision, deep learning models might overfit, and lack generalisation across different datasets or even different sensing modalities. This is especially problematic for multi-source data like SAR and multispectral data.

A further limitation is a lack of interpretability. The embeddings learned by deep networks are opaque and high-dimensional, and are not easily interpreted. This prevents broader use of these models in areas where interpretability is important.

1.5 Better Representations for Classification

To overcome the challenges of deep learning models, it is necessary to improve feature representation to support both learning performance and transparency. Rather than using continuous feature representations, discrete and interpretable representations can be used to enhance the efficiency and explainability of the learning process.

In this paper, a hybrid model is developed that uses a CNN to learn spatial features from satellite imagery, while using a Tsetlin Machine (TM) as a classifier. The convolutional neural network (CNN) can learn intricate spatial features, while the Tsetlin Machine can learn rules over binary feature representations.

A binary encoding method is proposed to transform the CNN features into binary representations suitable for the Tsetlin Machine. The encoding preserves relative feature patterns and spatial information, allowing the model to preserve information and enhance interpretability.

1.6 Remote Sensing Image Classification

Remote sensing image classification is the process of labeling regions of images according to their spectral and spatial features. Conventional methods usually approach this as a single-label classification, where a single label is assigned to each image or image patch.

But real-life scenes can be complex and include multiple land-cover classes in the same area. This makes the classification task more challenging, particularly at low resolutions. The performance of classification tasks is heavily reliant on the quality of the features as well as the model's capacity to learn local and global information.

The integration of deep feature learning with rule-based classification is a potential solution to enhance classification accuracy and interpretability.

1.7 Challenges

The key issues in satellite image classification are:

- **Low spatial resolution:** Relevant features such as textures and edges are lost, resulting in an ambiguous representation.
- **Mixed-pixel problem:** One pixel can contain several land-cover classes, complicating classification tasks.
- **Lack of training data:** Labelling satellite images is time-consuming and requires expertise.
- **Interpretability:** The predictions of deep learning models are not explainable.
- **Modal differences:** Sensors have different characteristics across different modalities.

1.8 Problem Statement

The goal of this study is to develop a satellite image classification system with high accuracy using limited labelled data, and which provides interpretable results.

Current approaches either emphasise accuracy through deep learning or interpretability through rule-based approaches, but seldom do both. A holistic approach is needed that offers both effective feature representation and interpretability, as well as limited supervision.

1.9 Key Contributions of the thesis

- An integrated CNN–Tsetlin Machine approach for interpretable classification
- An ordered binary encoding of CNN features
- A clause-based interpretability framework in line with model decisions
- Comprehensive experiments on various sensing under limited supervision

1.10 Thesis Organization

The remainder of this thesis is organised as follows. Chapter 2 presents a comprehensive literature review on satellite image classification and interpretable machine learning. Chapter 3 describes the datasets and experimental setup. Chapter 4 details the proposed methodology, including the CNN architecture, binary encoding process, and Tsetlin Machine integration. Chapter 5 presents the experimental results and discussion. Finally, Chapter 6 concludes the thesis and outlines future research directions and Chapter 7 gives an overview of the publications.

Chapter 2

Literature Review

2.1 Overview

This chapter provides an overview of research relevant to the proposed CNN-TM classification framework in five broad categories: satellite remote sensing and automated image classification challenges, traditional machine learning techniques, deep learning techniques, interpretable machine learning and explainable AI in remote sensing, and the Tsetlin Machine model and its variants. The survey reveals the conceptual and empirical gaps that justify the design of the proposed framework, and positions the work of this thesis in the wider field.

2.2 Satellite Remote Sensing: Platforms and Classification Issues

Satellite remote sensing comprises a family of sensors that vary in their measurement physics, spectral and spatial resolution and vulnerability to atmospheric and surface conditions [1]. Sensors that measure the visible-to-shortwave-infrared (VSWIR) spectrum of the Earth's surface offer spatially relevant information but are limited in their use by contamination from clouds and variability in illumination that can change spectral responses significantly between dates. Multispectral sensors, which collect data in four to twelve bands, are commonly used for land cover and crop-type mapping due to the compromise between spatial and spectral resolution that is achievable with this measurement technology. Hyperspectral sensors take this idea further, acquiring

hundreds of contiguous narrow bands to enable detailed discrimination of mineralogical and biochemical surface properties, but at the expense of high-dimensional features compared to the number of training samples [2].

Synthetic Aperture Radar (SAR) is a fundamentally different type of measurement: it uses coherent microwave pulses to measure the complex backscatter coefficient of surface targets, and is sensitive to the dielectric properties, surface roughness and geometrical structure of scatterers within each resolution cell [3]. This sensitivity makes SAR useful in combination with optical observations for applications such as soil moisture and forest biomass estimation and military target classification. But SAR images are corrupted by multiplicative speckle noise due to coherent interference of elementary scatterer returns, a noise process that has a statistical distribution that is different from the additive Gaussian noise implicit in most conventional neural network architectures [4].

The difficulties of automatic satellite image classification have been well documented. The Hughes effect, first observed in hyperspectral image classification, is used to describe the empirical fact that classification performance increases and then decreases with the number of spectral bands, once the number of training samples is exceeded [2]. This curse of dimensionality has led to significant effort in dimensionality reduction, feature selection and regularisation. In deep neural networks, a similar phenomenon occurs as overfitting when the network's capacity is significantly greater than the number of labelled samples, resulting in poor generalisation across acquisition conditions [5].

Another issue is the perceptual and spectral confusion between physically different classes under specific acquisition conditions. Rooftops and bare, dry soil have similar near-infrared reflectance. Different types of military vehicles can have similar amplitude signatures in SAR imagery when seen from specific angles where their primary scattering mechanisms converge. Different crop species at similar stages of development may have similar spectral signatures [6]. To overcome such intra-class and inter-class ambiguities, classifiers need to focus on local structural, contextual and multi-scale spatial information rather than global first-order statistics.

2.3 Traditional Machine Learning for Image Classification

2.3.1 Spectral Cues and Linear Classifiers

The first automated methods for satellite image classification were based on supervised parametric classifiers operating on engineered spectral features. Maximum likelihood and linear discriminant analysis (LDA) classifiers, and Mahalanobis distance measures were among the first practical methods, assuming Gaussian class-conditional probability distributions and estimating class statistics from training data [7]. These approaches were efficient but assumed Gaussian class-conditional distributions, were robust to model violations, and did not handle nonlinear spectral class boundaries typical for land cover classification.

Principal component analysis and linear discriminant analysis became popular as feature extraction preprocessors to tackle the Hughes phenomenon in hyperspectral classification. These methods reduced the effective dimensionality of spectral features while preserving class separability by projecting the high-dimensional spectral data into a lower-dimensional subspace of maximum variance (PCA) or maximum dissimilarity (LDA), at the expense of losing information in the lower variance principal axes.

2.3.2 Support Vector Machines

From the early 2000s, support vector machines (SVMs) were the preferred approach to hyperspectral and multispectral classification due to their theoretical foundation in structural risk minimisation, and their capacity to manage high-dimensional feature spaces using the kernel trick [7]. SVM classifiers are based on determining the maximum-margin hyperplane separating data in a kernel-induced feature space, and multiple comparative studies on benchmark data sets have verified their superior performance over parametric classifiers with constrained training data. The radial basis function kernel was particularly adept at modelling complex spectral distributions, and extensions to multiclass classification through one-versus-one and one-versus-all approaches enabled tackling multi-category land cover classification tasks.

Despite their ability to compete with other methods under limited training, there are practical concerns with SVMs in the current context. Appropriate kernel choice and hyperparameter optimization rely on expertise or costly cross-validation procedures, and SVMs are based on fixed, hand-engineered input feature descriptors that cannot be

adapted to particular tasks. Their effectiveness is therefore limited by the quality of the feature descriptor, a progressively limiting factor as the variety of satellite imaging techniques grows.

2.3.3 Ensembles and Random Forests

Random forests and gradient boosting ensembles became popular for satellite image classification after they were shown to be insensitive to irrelevant features, monotonic feature transformations, and provide feature importance measures. These characteristics made ensembles appealing in practice for diverse remote sensing data where the relevance of individual features was not clear in advance. Random forests achieved state-of-the-art performance on hyperspectral data, in combination with extended morphological attribute profiles that capture spatial context along with spectral information.

A key benefit of random forest classifiers is that they are inherently interpretable: individual decision trees are human-readable conditional rules, and the sum of feature importance scores (based on impurity reduction at tree nodes) gives quantitative estimates of per-feature relevance. But as the ensemble grows larger, it becomes difficult for humans to directly interpret the joint decision boundary and, like SVMs, ensemble methods rely on a fixed, non-updatable representation of input features, which cannot provide hierarchical abstraction of raw sensor data.

2.3.4 Sparse Coding and Dictionary Learning

Dictionary learning and sparse representation models provided an alternative mode of representation in which each image patch or spectral vector is represented as a sparse linear combination of atoms from a learned or fixed dictionary of basis vectors [6]. Sparse representational classifiers showed promising results on a range of hyperspectral datasets by leveraging the idea that samples from the same class lie in a low-dimensional subspace spanned by training samples from that class. Joint sparse representation generalised this idea to account for spatial neighbourhoods. While these approaches are conceptually appealing, they are computationally expensive at test time due to the need to solve an ℓ_1-regularised optimisation problem for each test sample, making them unsuitable for processing large image libraries.

2.4 Deep Learning for Satellite Image Classification

2.4.1 Convolutional Neural Networks

Convolutional neural networks (CNNs) have transformed the landscape of remote sensing image classification [7]. Whereas previous approaches to image recognition relied on hand-crafted feature extraction and classification pipelines, CNNs derive hierarchical features directly from the input data via end-to-end gradient descent. Initial models, such as AlexNet, VGGNet and ResNet, were transferred to multiclass classification of multispectral and hyperspectral imagery, achieving significant gains in accuracy on standard benchmark datasets, compared to classical approaches.

CNNs are naturally adapted to the structure of optical satellite images, where the discriminative information is encoded in local texture and edge patterns, as well as the spatial organisation of surface objects at multiple scales. Multi-scale CNN models that simultaneously operate on different scales of input imagery have been found to enhance classification performance of complex scenes containing objects of different sizes. Residual, dense skip connections, and channel attention blocks have also been used to enhance the expressiveness and stability of deep CNN-based satellite image classifiers.

2.4.2 Recurrent and Attention-Based Architectures

Recurrent neural networks (RNNs), especially long short-term memory (LSTM) networks, were also investigated as spectral feature sequence learners in the context of hyperspectral classification, by considering the spectral response of a pixel as a time series of band-wise measurements [2]. This enabled LSTMs to capture long-range spectral dependencies that are challenging to learn from local convolutional features. Attention, first introduced in natural language processing, was then introduced into spectral as well as spatial classification to allow for selective and data-dependent weighting of spectral bands or spatial pixels.

2.4.3 Vision Transformers and Self-Attention

Vision Transformers (ViTs) and their variants for remote sensing have gained significant attention after achieving impressive performance on natural image benchmarks [8]. ViT models divide images into non-overlapping patches and process them as tokens using

multi-head self-attention, enabling modelling of global spatial contexts that are challenging to capture with locally receptive CNNs. Spectral Transformers extended this idea to hyperspectral imagery by using spectral bands or band subsets as tokens, modelling global spectral interactions. Although transformers have shown promising accuracy, they require vast pre-training data and extensive computational resources, and are therefore less suited for direct application to data-limited remote sensing data.

2.4.4 Few-Shot and Meta-Learning Methods

The lack of labelled satellite data has fostered considerable research on few-shot learning approaches to attain sufficient classification accuracy with very little labelled data available [5]. Prototypical networks, model-agnostic meta-learning (MAML) and relation networks have been successfully applied to remote sensing classification problems, with adjustments to accommodate multispectral and multi-resolution satellite data. Meta-learning strategies to different extents tackle sample inefficiency, but they often require large meta-training datasets from related tasks, which may be unavailable for domain-specific sensor modalities.

2.4.5 Multimodal Fusion Architectures

The emergence of complementary sensing modalities - optical, SAR, hyperspectral, and LiDAR - has prompted the development of multimodal fusion approaches that learn multimodal representations from diverse sensor data [9, 10]. Early fusion methods combine different sensor inputs at the feature level and feed them to a common network. Late fusion schemes learn separate representations for each sensor and then fuse predictions. Cross-sensor attention modules adaptively weight the different sensor inputs based on the input, providing more flexibility than hard-coded fusion rules. Multimodal fusion can provide significant performance improvements but adds complexity and necessitates multi-sensor synchronisation, which may not be readily accessible in practice.

2.4.6 Interpretability of Machine Learning and Explainable AI in Remote Sensing

2.4.7 The Interpretability Imperative

The increasing use of machine learning models in high-impact remote sensing use cases - such as damage detection for critical infrastructure, crop yield prediction, and military target classification - has led to a rise in the need for interpretability beyond mere predictive performance [11]. Interpretability provides multiple practical benefits: it helps to verify that a model is using physically meaningful features rather than artefacts of the data; it can be used to satisfy regulatory requirements in applications where automated decisions must be auditable; and it can be used to identify failure modes and out-of-distribution samples that might affect model reliability.

The machine learning literature divides models into those that are inherently interpretable (e.g. decision trees, linear methods, and rule-based models) and those for which interpretability is achieved through post-hoc methods that are applied after a model is trained. For remote sensing use cases where spatial attribution maps - i.e. maps of the regions in an image that contributed to the classification decision - are of particular interest, both types of methods have been investigated.

2.4.8 Post-Hoc Explanation Methods

GradCAM and its derivatives generate spatial attribution maps by backpropagating the gradient of the class score (the output of the CNN) with respect to the intermediate CNN feature maps, scaled by their global average pooling weights [11]. These approaches are commonly used in remote sensing to determine the image regions that a trained CNN focuses on when making a decision. LIME fits a locally faithful linear surrogate model to the predictions of a trained model for a given input, by sampling a neighbourhood of inputs around that input [12].

A key weakness of all post-hoc explanation techniques is formal disentanglement from the prediction process. Explanation is generated by a different computational mechanism than the model itself, which merely mimics the behaviour of the model but does not capture the mechanisms that shaped the prediction. As a result, the features highlighted in a GradCAM map may not be the features that caused the classification decision, and the local linear approximation of a LIME explanation may be a poor fit to the local decision

boundary. In the context of operational remote sensing systems, where attribution maps are used to explain and justify decisions to human users or regulatory agencies, this lack of fidelity can be a critical issue.

2.4.9 Intrinsically Interpretable Classifiers

Decision trees, rule sets and fuzzy inference systems are the traditional class of inherently interpretable classifiers, which make predictions as the logical outcome of explicit if-then rules that can be scrutinised and verified by domain experts [7]. Their representational power is however significantly weaker than deep neural networks and they have historically achieved much lower performance on high-dimensional satellite image classification tasks compared to CNNs. The possibility of marrying deep feature extractors and shallow rule classifiers to balance the representational capacity and interpretability of the former and latter, respectively, has therefore gained attention.

2.5 The Tsetlin Machine

2.5.1 Model and Theoretical Background

The Tsetlin Machine (TM) is a propositional rule learning model that uses Tsetlin automata, a type of finite-state reinforcement learning machine which learns to include or exclude Boolean literals from conjunctive clauses via a stochastic feedback scheme. A clause is a conjunction of positive and negated Boolean literals, and classification is based on the weighted sum of the clause outputs, with the weights of the clauses being updated by a feedback mechanism that combines generalisation and memorisation through the use of a sparsity hyperparameter s that regulates the level of detail in learned patterns.

The key architectural feature of the TM is that its predictions are direct deterministic logical consequences of the set of trained clauses: for every input and clause configuration, the class predicted is a logical consequence of the conjunction of the active literals in each contributing clause, which in turn are consequences of the input features that triggered the clauses [13]. This is structural rather than approximate transparency, since it is not derived from a post-hoc analysis, and is exact rather than probabilistic. This aspect justifies its use as the classifier in the CNN-TM pipeline proposed here, where it facilitates attribution heatmaps based on the same clause activations that were used to make the prediction.

2.5.2 Extensions and Applications

The Tsetlin Machine has been extended in a number of ways since its inception. The Convolutional Tsetlin Machine reuses clause patterns across spatial locations, akin to weight sharing in convolutional neural networks, to efficiently classify spatially structured image data. Multi-granularity variants include multiple pattern specificity levels using dynamic thresholding. Regression extensions can predict continuous values. It has been applied to image classification, sentiment analysis, anomaly detection and natural language processing, with generally good accuracy, much smaller model sizes, and better interpretability than neural network approaches.

There are relatively few applications of the Tsetlin Machine to remote sensing. A few studies have investigated classification of hyperspectral data by the TM using spectral features, with promising results for limited training budgets. However, the problem of adapting the TM to spatially structured multi-modal satellite imagery without modality-specific feature engineering is yet to be tackled. This need underpins the CNN front-end and binary encoding module presented in this thesis, which produce modality-independent Boolean signatures that can be used with the TM for classification across various sensing modalities.

2.6 Insights from the Literature Review

Our survey of the literature on satellite image classification suggests a number of insights into the current state of different approaches. These observations aid in understanding the advances that have been made, as well as the challenges that remain.

2.6.1 Classical Machine Learning

Traditional machine learning algorithms like Support Vector Machines and Random Forests have been extensively applied for satellite image classification, especially in the context of hyperspectral and multispectral imagery. These methods perform well with a limited amount of labelled data and are somewhat interpretable because of their decision-making structure [7].

But their success is heavily reliant on hand-crafted features. With the growing complexity and modality of satellite data, manual feature extraction is often insufficient to represent

the complex patterns. This restrains scalability and flexibility of classical methods in practical applications.

2.6.2 Deep Learning Approaches

Deep learning approaches, especially Convolutional Neural Networks, have greatly enhanced classification accuracy by extracting hierarchical features directly from the data [7]. Variants that introduce attention, residual connections and multi-scale features have increased their ability to capture complex spatial and spectral interactions.

Vision Transformers take this a step further by modelling long-range dependencies with attention mechanisms [8]. Likewise, few-shot learning methods seek to alleviate data scarcity issues by enhancing sample efficiency [5].

However, deep learning models still require large datasets with labels, and are computationally costly. More crucially, they are inherently non-interpretable and thus unsuitable for critical applications.

2.6.3 Explainable AI Techniques

To make deep learning models more interpretable, post-hoc explanation techniques like GradCAM and LIME are extensively used [11, 12]. These techniques generate explanations in the form of visual highlights on the input image.

But these explanations are not intrinsic to the black box model and are obtained via approximation procedures. This can result in explanations that do not reflect the true decision-making process of the model, and may be unreliable in real-world settings.

2.6.4 Tsetlin Machine-Based Learning

The Tsetlin Machine provides an alternative learning mechanism that learns logical rules over binary features [13]. It is inherently interpretable, with predictions being directly explainable by logical clauses.

Moreover, Tsetlin Machines are known to achieve good performance with relatively little data, so they are a suitable choice in remote sensing scenarios where data is scarce. Yet, their use is constrained with image data due to the binary nature of the inputs and lack of an efficient feature extraction strategy for complex spatial data.

2.6.5 Comparative Analysis

From the above discussion, we provide a comparison of the reviewed approaches in Table 2.1.

TABLE 2.1: Comparative summary of reviewed methods

Method	**Key Idea**	**Strengths**	**Limitations**
Traditional ML (SVM, RF) [7]	Utilises handcrafted features with statistical or kernel-based classifiers	Effective under limited data; relatively interpretable; computationally efficient	Limited representational capacity; depends on feature engineering
CNN-based Models [7]	Learns hierarchical spatial features directly from raw data	High accuracy; captures complex spatial patterns	Requires large labelled datasets; lacks interpretability
Transformers [8]	Employs self-attention to model global dependencies	Captures long-range relationships effectively	Computationally intensive; data-demanding
Few-shot Learning [5]	Learns transferable representations from limited samples using meta-learning	Improves sample efficiency	Performance depends on meta-training; limited robustness across domains
Explainable AI [11, 12]	Provides post-hoc explanations of model predictions	Enhances interpretability of black-box models	Explanations are approximate and not inherently linked to the model
Tsetlin Machine [13]	Learns logical rules over Boolean feature representations	Intrinsically interpretable; effective in low-data regimes	Requires binary inputs; lacks native feature extraction capability

2.7 Research Gap

Our review of the current literature raises the question of a trade-off between performance, interpretability, and efficiency in satellite image classification.

Traditional machine learning systems are interpretable and perform well with small amounts of data, but rely on pre-determined features which limits their capacity to learn from complex, high-dimensional data. Deep learning approaches address this issue by automatically learning effective feature representations, which leads to improved

classification accuracy. But they require a large amount of labelled data and are not interpretable.

Visual explanations of deep learning models are offered by post-hoc explainability methods, which seek to address this issue. However, this approach doesn't necessarily produce faithful explanations, as it is not inherently part of the model's training.

The Tsetlin Machine is a promising alternative in this regard, as it offers inherently interpretable rule-based learning that is efficient with little data. But its use on satellite data is limited by the lack of an efficient feature extraction method and the need for binary data representations.

These insights point to a significant gap in the literature: there is no single approach that simultaneously delivers high accuracy, efficiency with limited data, as well as interpretability.

This presents an opportunity for the proposed approach in this thesis, combining convolutional neural networks (CNN) for feature learning with the Tsetlin Machine (TM) for interpretable classification. By leveraging these respective advantages, the proposed CNN-Tsetlin Machine approach strives to provide an accurate and faithful classification system that is also data efficient.

Chapter 3

Proposed Methodology

The approach contains three stages: a multi-scale CNN learns spatial feature maps from raw SAR amplitude images; a structured binary encoding module transforms them into binary vectors; and a TM that finds propositional classification rules over binary features. Then an interpretation module projects the to image areas without using gradients. Our entire pipeline is shown in Fig. 3.1.

Design Rationale. Our proposed CNN-TM model aims to solve the following three key problems in remote sensing image classification: (i) learning effective spatial representations from complex images, (ii) handling a lack of labelled data, and (iii) offering interpretable results.

Convolutional neural networks (CNNs) excel at feature learning, but are not interpretable. On the other hand, interpretable rule-based models like the Tsetlin Machine (TM) rely on binary structured inputs. This approach integrates multi-scaleCNN feature extraction and structured binary encoding, allowing for learning of logical rules over meaningful features.

3.0.1 Problem Formulation

We formalise the classification problem addressed by the proposed framework.

Given a dataset

$$\mathcal{D} = \{(I^{(n)}, y^{(n)})\}_{n=1}^{N},$$

where $I^{(n)} \in \mathbb{R}^{H \times W \times C}$ denotes a satellite image and $y^{(n)} \in \{1, \ldots, K\}$ is the corresponding class label, the objective is to learn a mapping:

$$\hat{y}^{(n)} = f_\theta\big(b^{(n)}\big)$$

where the binary representation $b^{(n)}$ is obtained through a two-stage transformation:

$$Z^{(n)} = g_\psi(I^{(n)}), \quad b^{(n)} = \phi(Z^{(n)})$$

Here, g_ψ denotes the multi-scale CNN feature extractor producing a feature map

$$Z^{(n)} \in \mathbb{R}^{C\times H\times W},$$

ϕ represents the structured binary encoding function that converts $Z^{(n)}$ into a Boolean vector

$$b^{(n)} \in \{0,1\}^d,$$

and f_θ is the Tsetlin Machine classifier that operates on binary inputs.

The objective is to achieve accurate, robust, and interpretable classification under limited supervision.

3.0.2 Input Preprocessing

Each image is resized to 64×64 pixels and linearly normalised to [0, 1]. Because the multi-scale front-end expects a three-channel input, the single-channel SAR amplitude image is replicated across three channels before processing. No information is introduced or lost by this replication; the three parallel filter banks develop independent responses to the same spatial content.

3.0.3 Multi-Scale CNN Feature Extractor

We apply the same pre-processing to all datasets. Data augmentation is not applied, nor has the network been pretrained on other data, so that the network's performance can be evaluated based on its inherent learning ability with limited supervision.

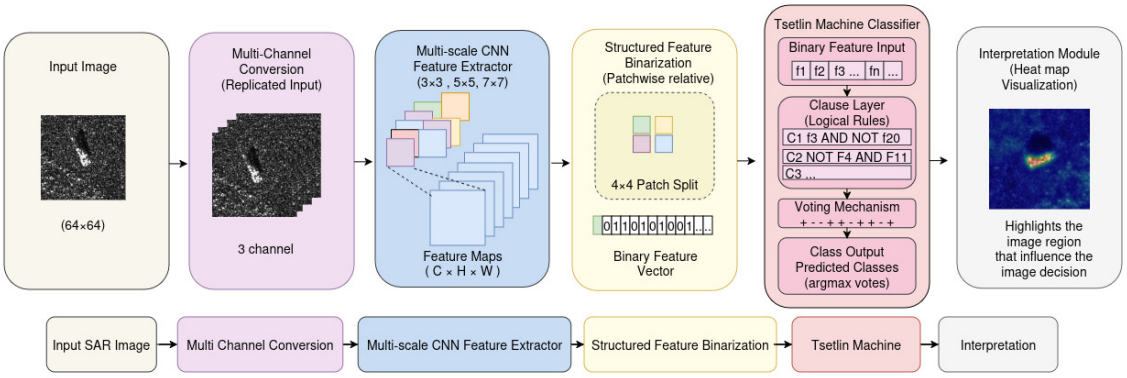


FIGURE 3.1: End-to-end CNN–TM classification and interpretation pipeline. The input SAR image is replicated to three channels, processed by the multi-scale CNN extractor, transformed into a structured binary representation via patch-wise relative binarisation, classified by the TM, and finally attributed spatially by the clause-driven interpretation module.

3.0.3.1 Architecture

The network has three parallel convolutional paths with 3×3, 5×5, and 7×7 are applied directly to the input, zeroed to maintain spatial resolution, and the responses are combined channel-wise in a volume of depth 48. This is then followed by four blocks of Conv→BatchNorm→ReLU, with 2×2 max-pooling after the first three blocks, reducing spatial resolution by half and increasing depth at each step. The last block has size 8×8 without pooling and employs dropout with $p = 0.4$. The output is a feature volume $\mathbf{Z} \in \mathbb{R}^{C \times H \times W}$ with $C = 256$, $H = W = 8$, as illustrated in Fig. 3.2.

The three different kernel sizes span different frequency scales edges: 3×3 for fine structure edges, 5×5 resolves medium scale texture, and 7×7 averages coarse intensity changes due to shadows and specular lobes.

3.0.3.2 Training Configuration

The network is trained end-to-end using the Adam optimiser [14] with a learning rate of 10^{-3}, weight decay of 10^{-4}, and mini-batch size of 32, for 50 epochs. Cross-entropy loss with label smoothing ($\varepsilon = 0.1$) is used to improve calibration under the small sample budget. An auxiliary linear classification head $\mathbf{W}_{\text{cls}}$ is attached during training and discarded at inference, where classification is handled entirely by the TM. Dropout at $p = 0.4$ discourages memorisation of sample-specific speckle patterns under the 121-sample-per-class constraint.

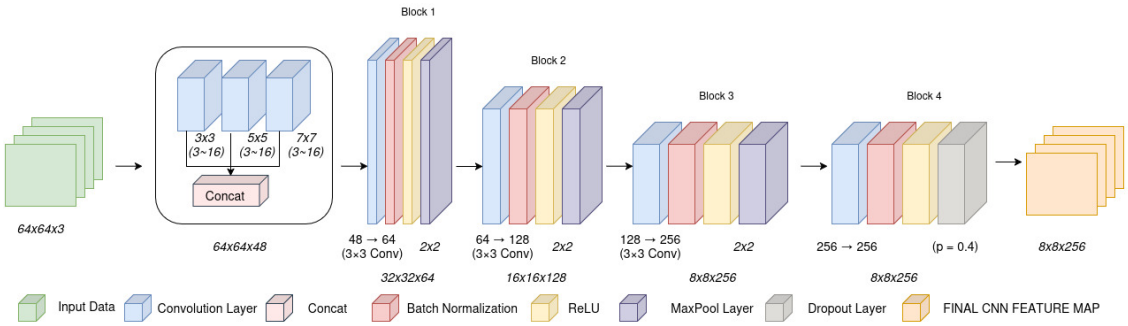


FIGURE 3.2: Proposed multi-scale CNN architecture. Parallel 3×3, 5×5, and 7×7 convolutions capture multi-scale features, followed by concatenation and four Conv-BN-ReLU blocks with progressive max-pooling. A final dropout layer ($p = 0.4$) improves generalisation, producing an 8×8×256 feature representation.

Implementation Details. The CNN is implemented using PyTorch and trained on a GPU for efficient optimization. The feature map resolution of 8×8 is not manually tuned but emerges from three successive pooling operations applied to the fixed input size of 64×64.

The auxiliary classification head is used only during training to stabilise feature learning and is removed during inference, ensuring that final predictions depend entirely on the Tsetlin Machine.

3.0.4 Structured Binary Feature Encoding

The encoding module converts $\mathbf{Z}$ into a high-dimensional binary vector while preserving three physically meaningful properties of the SAR response: relative intensity, texture heterogeneity, and peak activation strength.

3.0.4.1 Spatial Partitioning

Each channel $\mathbf{Z}^c \in \mathbb{R}^{8\times 8}$ is partitioned into a non-overlapping 4×4 grid of 16 spatial patches $\mathbf{P}^c_{ij}$, $i, j \in \{0, \ldots, 3\}$, each of size 2×2 pixels. This resolution conveys spatial context—distinguishing front/rear and left/right regions of the vehicle image—without reducing patches to a size too small for reliable statistical estimation, as illustrated in Fig. 3.3.

3.0.4.2 Relative Normalisation

For each channel c the mean over the channel, μ_c and deviation σ_c over the 64 spatial locations. The statistics of each patch are then calculated relative to references:

$$\mu_{\text{rel}} = \frac{\mu_{ij}}{\mu_c + \varepsilon}, \tag{3.1}$$

$$\sigma_{\text{rel}} = \frac{\sigma_{ij}}{\sigma_c + \varepsilon}, \tag{3.2}$$

$$\text{cv} = \frac{\sigma_{ij}}{\mu_{ij} + \varepsilon}, \tag{3.3}$$

where μ_{ij}, σ_{ij}, and m_{ij} denote the mean, standard deviation and maximum activation of $\mathbf{P}^c_{ij}$, and $\varepsilon = 10^{-6}$ prevents division by zero. Relative normalisation makes all encoded features invariant to the channel-wise scaling differences of the three parallel front-end branches.

In practice, μ_{rel} is an indicator of localised presence; σ_{rel} identifies specular (homogeneous) scattering (heterogeneous) regions; and cv enhances intensity edges in low backgrounds, including dihedral and dihedral and shadow-edge responses. The encoding approach retains spatial locality while facilitating binarisation. Local binarisation preserves spatial context, enabling the Tsetlin Machine to learn spatial rules.

Relative normalisation normalises the scaling of different channels, and binarisation using multiple thresholds encodes ordinal information of feature responses without increasing the dimensionality significantly.

3.0.4.3 Multi-Threshold Binarization

Each descriptor is quantized using multiple thresholds, encoding ordinal information directly in the bit pattern. The nine binary features per patch are defined in Table 3.1: three threshold levels are applied to μ_{rel} (bits f_1–f_3), two to σ_{rel} (bits f_4–f_5), two to cv (bits f_6–f_7), and two to the patch maximum m_{ij} (bits f_8–f_9). The thresholds for m_{ij} are set adaptively at μ_c and $\mu_c + \sigma_c$, scaling with each channel's activation statistics. Each patch therefore produces a 9-bit vector; concatenating all 16 patches across all $C = 256$ channels yields a binary feature vector of dimension $256 \times 16 \times 9 = 36{,}864$ bits per sample.

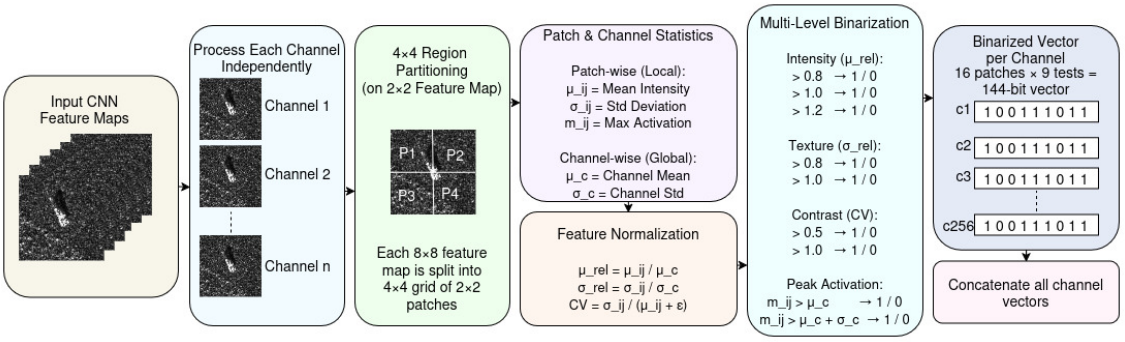


FIGURE 3.3: Structured binary feature encoding. Each 8×8 CNN feature-map channel is divided into a 4×4 grid of non-overlapping 2×2 patches (16 patches per channel). Per-patch and channel-level statistics are computed and channel-normalised to yield the three relative descriptors (μ_{rel}, σ_{rel}, cv). Multi-threshold binarisation produces nine Boolean bits per patch; concatenation across all 256 channels yields a 36,864-bit feature vector.

Algorithm 1 Multi-Scale CNN Feature Extraction

Require: Dataset $\mathcal{D} = \{(\mathbf{I}_n, y_n)\}_{n=1}^{N}$, CNN parameters $\boldsymbol{\theta}$
Ensure: Feature maps $\{\mathbf{Z}_n\}$

```
for each image I_n ∈ D do
    I_n ← Resize(I_n, 64×64)
    I_n ← I_n/255
    X_n ← ReplicateChannels(I_n)                    ▷ 64×64×3
end for
Initialise f_θ with parallel 3×3, 5×5, 7×7 front-end branches
for epoch = 1 to 50 do
    for each mini-batch (X, y) of size 32 do
        F_3 ← ReLU(Conv_{3×3}(X))
        F_5 ← ReLU(Conv_{5×5}(X))
        F_7 ← ReLU(Conv_{7×7}(X))
        F ← Concat(F_3, F_5, F_7)
        Z ← f_θ(F)                  ▷ 4 Conv-BN-ReLU blocks; ×Pool on first 3
        p ← AdaptiveAvgPool(Z)
        ŷ ← W_cls p
        L ← CrossEntropy_{ε=0.1}(y, ŷ)
        Update θ, W_cls via Adam (lr = 10^-3, wd= 10^-4)
    end for
end for
for each sample X_n do
    Z_n ← f_θ(X_n)                  ▷ W_cls discarded at inference
end for
return {Z_n}
```

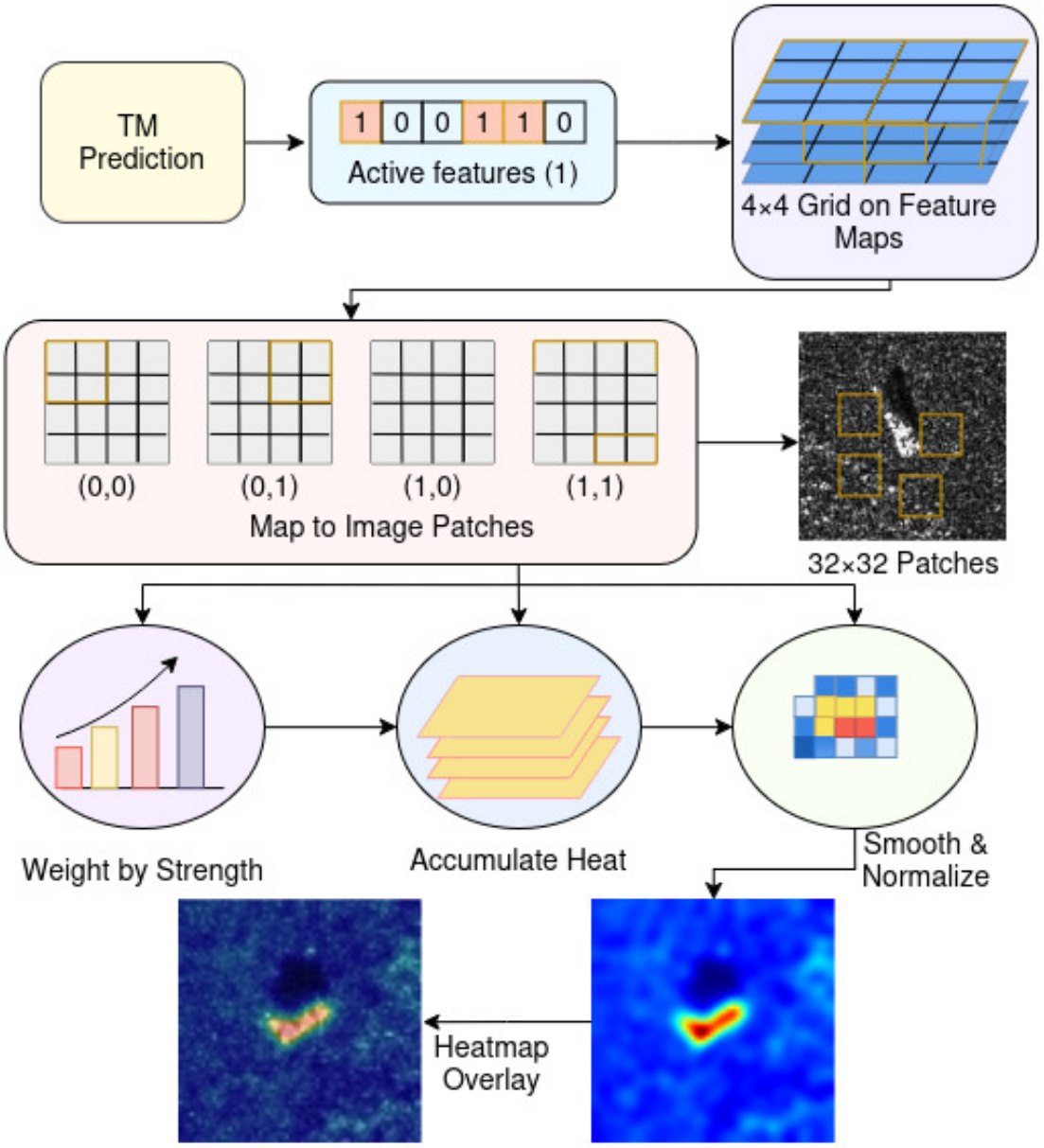


FIGURE 3.4: Clause-driven interpretation. Activated binary features are spatially localised via feature-map indexing and accumulated with type-dependent weights to construct a high-resolution heatmap. Gaussian smoothing and normalisation produce the final attribution visualisation, which is structurally consistent with the TM's active clause set.

3.0.5 Tsetlin Machine Classifier

A Tsetlin Machine (TM) [13] is a rule-based learning model that uses propositional logic expressions Boolean input features. In our approach, the TM is applied to the binary features extracted from the structured representation of CNN feature maps.

TABLE 3.1: Binary Feature Encoding: Conditions and SAR Interpretation

Bit	**Condition**	**Descriptor**	**SAR Meaning**
f_1	$\mu_{\text{rel}} > 0.8$	Intensity	Mildly above channel mean
f_2	$\mu_{\text{rel}} > 1.0$	Intensity	Clearly above mean
f_3	$\mu_{\text{rel}} > 1.2$	Intensity	Dominant scattering centre
f_4	$\sigma_{\text{rel}} > 0.8$	Texture	Mildly heterogeneous
f_5	$\sigma_{\text{rel}} > 1.0$	Texture	Multi-bounce or clutter
f_6	$\text{cv} > 0.5$	Contrast	Distributed scattering
f_7	$\text{cv} > 1.0$	Contrast	Dihedral or shadow edge
f_8	$m_{ij} > \mu_c$	Peak	Notable point scatterer
f_9	$m_{ij} > \mu_c + \sigma_c$	Peak	Strong point scatterer

Each class is defined by a conjunction of clauses, each clause being conjunction of Boolean features (or their negations). These clauses create a decision function that "votes" for or against each class. The class with the most votes wins the prediction. the highest vote.

In contrast to gradient-based approaches, the TM is based on a set of stochastic finite-state machines called Tsetlin Automata. Each automaton individually learns to include or exclude a literal in a clause. The training is based on positive and negative feedback from the results of classification, allowing the model to learn interpretable logical rules.

Each class has two sets of clauses:

- **Positive clauses:** Vote in favour of the class
- **Negative clauses:** Vote against the class

The final decision is based on the difference between the positive decision score. This allows a clear connection between between features and the prediction.

The TM has three important hyperparameters:

- **Clauses ($C_{\mathbf{TM}}$):** Controls model capacity. A higher number of clauses enables the model to learn more intricate patterns but at the expense of higher and chance of overfitting.
- **Threshold (T):** Bounds the range of the vote of the clauses to $[-T, +T]$. This enforces consensus among clauses and prevents individual rules from dominating the prediction.

- **Specificity (s):** Controls the likelihood of a literal being included in a rule. Higher values lead to more specific rules, lower values lead to generalisation.

In this paper, the TM is trained for 50 epochs with stochastic Tsetlin Automata updates executed on GPU. Weighted clauses are used to allow to weight highly confident rules, and feature negation allows modelling of both the presence and absence of patterns, which is which is significant in SAR images.

3.0.6 Model Hyperparameter Configuration

TABLE 3.2: Hyperparameter Settings of the Proposed CNN–TM (Aligned with Table **??**)

Component	**Parameter Setting**
CNN	
Kernel Sizes (k_1, k_2, k_3)	3, 5, 7
Output Feature Map	$256 \times 8 \times 8$
Dropout (p)	0.4
Learning Rate (η)	3×10^{-4}
Weight Decay (λ)	1×10^{-4}
Batch Size (B_b)	32
CNN Epochs (E_{cnn})	50
Label Smoothing (ε_{ls})	0.1
Binary Encoder	
Grid Size (G)	4
Bits per Patch (B)	9
Peak Weight (w_p)	2.0
Base Weight (w_b)	1.0
Stability Constant (ε)	10^{-6}
Tsetlin Machine	
Number of Clauses (C_{TM})	4000
Threshold (T)	2500
Specificity (ρ)	6.0
TM Epochs (E_{tm})	50

The hyperparameters of the model are divided into three components corresponding to the CNN front-end, the structured binary encoding stage, and the Tsetlin Machine classifier. The CNN hyperparameters define the feature extraction process. Multi-scale convolutional kernels $(3, 5, 7)$ are employed to capture spatial patterns at different receptive field sizes, enabling robust representation of both fine-grained and coarse structures. The resulting feature map of size $256 \times 8 \times 8$ is regularised using dropout ($p = 0.4$) to mitigate overfitting in data-scarce regimes. Training stability is further ensured through a controlled learning rate ($\eta = 3 \times 10^{-4}$), weight decay ($\lambda = 10^{-4}$),

batch size of 32, and label smoothing ($\varepsilon_{\text{ls}} = 0.1$), with the network trained for 50 epochs. The binary encoding hyperparameters govern the transformation of continuous CNN features into structured propositional representations. The feature map is partitioned into a fixed grid ($G = 4$), and each patch is encoded using $B = 9$ bits to capture intensity variations, local texture, and salient peak responses. A weighted encoding scheme is adopted, where peak activations are emphasised ($w_p = 2.0$) relative to baseline responses ($w_b = 1.0$), improving discriminative focus on informative regions. A small stability constant ($\varepsilon = 10^{-6}$) is introduced to ensure numerical robustness during encoding. The Tsetlin Machine hyperparameters control the logical learning process and model complexity. The number of clauses ($C_{\text{TM}} = 4000$) determines the representational capacity of the model, while the threshold ($T = 2500$) regulates the voting mechanism among clauses. The specificity parameter ($\rho = 6.0$) balances clause granularity, influencing how general or specific the learned patterns are. The Tsetlin Machine is trained for 50 epochs to ensure convergence. Unless otherwise stated, this configuration is consistently applied across all datasets to ensure a fair and controlled comparison, with only selected Tsetlin Machine parameters tuned independently where required.

3.0.7 Hyperparameter Tuning of Tsetlin Machine

To achieve the best performance across the different sensing sensing, we optimise the TM hyperparameters with the Optuna library using a Tree-structured Parzen Estimator (TPE) sampler. Compared to random and grid search methods, Optuna offers adaptive sampling, enabling efficient search of the hyperparameter space.

A total of **100 trials** are run for each dataset. In each trial, the TM is trained on the binary representation of the dataset and measured on classification accuracy.

3.0.7.1 Search Space

We consider the following hyperparameter space:

Table 3.3: Search space for the hyperparameters of Tsetlin Machine

Parameter	Range	Type
Clauses (C_{TM})	2000 – 8000	Step = 500
Threshold (T)	100 – 2000	Step = 100
Specificity (s)	2.0 – 8.0	Continuous

3.0.7.2 Optimisation Results

Table 3.4 shows some of the evaluated configurations for MSTAR-10.

Table 3.4: Example Optuna trials for MSTAR-10

Trial	Clauses	T	Accuracy (%)
1	3000	500	78.42
2	4500	800	82.15
3	6000	1200	85.73
4	4000	2500	**98.39**
5	7500	1600	86.90

The findings demonstrate that more clauses leads to performance until reaching a saturation point after which the improvement is reduced.

3.0.7.3 Chosen Hyperparameters per Dataset

Table 3.5: Optuna selected hyperparameters

Dataset	Clauses	T	s
Indian Pines	6500	1000	2.156
SAMPLE-10	7500	300	7.761
MSTAR-10	4000	2500	6.0
EuroSAT	3800	1800	4.0

The diversity in best parameters is due to the nature of characteristics:

- SAR data benefits from a greater number of clauses to structural variations.
- Hyperspectral data benefits from lower specificity to generalisation.
- Simulated data may require higher specificity to capture fine-grained patterns.

3.0.7.4 Discussion

The optimization process reveals that:

- Number of clauses is the most important factor.
- Threshold helps by requiring clauses to agree.
- Specificity regulates overfitting and generalization.

The optimization curves (see Appendix) show an initial improvement space, suggesting that the TM hyperparameter space is very appropriate for Bayesian optimization.

In summary, the chosen hyperparameters are an almost ideal compromise between capacity, stability and generalization. datasets.

3.0.8 Interpretation Module

The predictions of TM are conjunctions of Boolean literals, can be traced back to the input features. For test sample $\mathbf{x}_n$ is:

$$\mathcal{A}_n = \{k \mid \mathbf{x}_n[k] = 1\}. \tag{3.4}$$

Each $k \in \mathcal{A}_n$ corresponds to a channel c_k and patch (i_k, j_k) in $\mathbf{Z}$. The corresponding normalized patch $\mathbf{R}_k$ is added to a heatmap $\mathbf{H}_n$, initialized to zero:

$$\mathbf{H}_n \mathrel{+}= w_k \cdot \frac{\mathbf{R}_k}{\max(\mathbf{R}_k) + \varepsilon}, \quad \forall\, k \in \mathcal{A}_n, \tag{3.5}$$

with the accumulation weight:

$$w_k = \begin{cases} 2.0 & \text{if } k \in \{f_8,\, f_9\}, \\ 1.0 & \text{otherwise.} \end{cases} \tag{3.6}$$

Peak-activation features $(f_8,\, f_9)$ are up-weighted to equalise their total energy against the seven mean- and bits per patch; otherwise the diffuse patterns would dominate features would overpower scatterer localisation. The heatmap is then Gaussian-smoothed ($\sigma = 1.5$ pixels), bilinearly upsampled to 64×64, normalised to [0, 1], and overlaid on the original image as shown in Fig. 3.4. Importantly, $\mathbf{H}_n$ is from the same active set of used for prediction - no proxy model or inference step pass is required.

The new interpretation mechanism is inherent to the model's decision-making process, unlike post-hoc interpretability methods like GradCAM or LIME. The heatmaps are generated from active clauses of the Tsetlin Machine, allowing faithful explanations without the need for surrogate models or gradient-based methods.

Algorithm 1 details multi-scale CNN feature extraction and Algorithm 2 details binary encoding, TM training, interpretation.

The CNN-TM approach combines deep learning and rule-based learning for interpretable satellite image classification.

The CNN extracts spatial features at multiple scales from input images, highlighting local texture and global patterns. The structured binary encoding produces a compact binary representation of the features while maintaining spatial context. The Tsetlin Machine learns meaningful logical rules over these features, allowing for interpretable and explainable decision-making.

This integration overcomes the shortcomings of current methods by balancing performance, interpretability, and efficiency, enabling its application in real-world remote sensing scenarios.

Algorithm 2 Binary Encoding, TM Training, and Interpretation

Require: Feature maps $\{\mathbf{Z}_n\}$, labels $\{y_n\}$, $C_{\text{TM}} = 4000$, $T = 2500$, $s = 6.0$
Ensure: Predictions $\hat{y}$, heatmaps $\{\mathbf{H}_n\}$

```
for each sample Z_n do
    b_n ← ∅
    for each channel c in Z_n do
        μ_c ← MEAN(Z^c)
        σ_c ← STD(Z^c) + ε
        Partition Z^c into 4 × 4 patches P^c_ij
        for each patch P^c_ij do
            μ_ij ← MEAN(P^c_ij)
            σ_ij ← STD(P^c_ij)
            m_ij ← MAX(P^c_ij)
            μ_rel ← μ_ij/(μ_c + ε)
            σ_rel ← σ_ij/(σ_c + ε)
            cv ← σ_ij/(μ_ij + ε)
            Compute f_1, ..., f_9
            b_n ← b_n ⊕ [f_1, ..., f_9]
        end for
    end for
end for
Initialise TM(C_TM, T, s)
for epoch = 1 to 50 do
    FIT(b_tr, y_tr)
end for
ŷ ← PREDICT(b_te)
for each test sample b_n do
    H_n ← 0
    A_n ← {k | b_n[k] = 1}
    for each k ∈ A_n do
        Retrieve R_k
        R_k ← R_k/(max(R_k) + ε)
        if k ∈ {f_8, f_9} then
            w_k ← 2.0
        else
            w_k ← 1.0
        end if
        H_n ← H_n + w_k R_k
    end for
    H_n ← GAUSSIANSMOOTH(H_n, 1.5)
end for
return ŷ, {H_n}
```

Chapter 4

Dataset Description and Experimental Protocol

This chapter introduces the datasets and common experimental protocol used in this work. The datasets encompass a variety of sensor types, such as Synthetic Aperture Radar (SAR), multispectral optical, simulated SAR and hyperspectral data. This allows us to evaluate the generalisation of our models under a range of different physical sensing conditions.

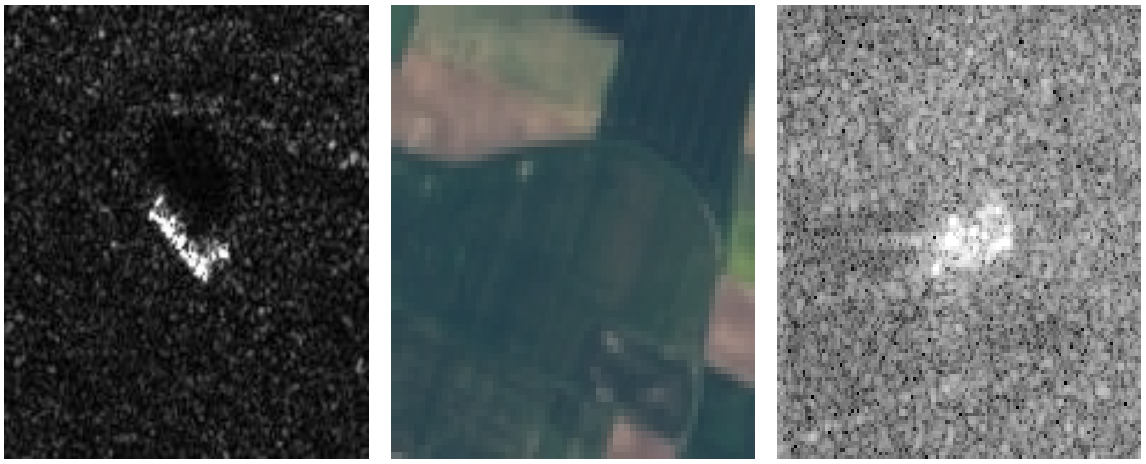
We adhere to a stringent training budget of 121 samples per class for all datasets, akin to remote sensing applications with scarce labelled samples. Images are cropped to 64×64 pixels and normalised to $[0, 1]$. The datasets are summarised in Table 4.1.

4.1 MSTAR-10 Dataset

The MSTAR dataset [15] is a popular Automatic Target Recognition (ATR) benchmark for SAR. It is acquired with an X-band spotlight SAR sensor and it contains ground vehicle targets acquired with different viewpoints.

We use a ten-class subset of MSTAR, involving a variety of vehicle types in different configurations, orientations, and depression angles. The training is done with 121 samples per class at a depression angle of 17° and the testing is done with images at a depression angle of 15°, which provides 2,426 test samples.

The cross-depression test results in a distribution shift, turning MSTAR into a difficult dataset for generalisation evaluation.



(a) MSTAR-10 dataset (b) EuroSAT dataset (c) SAMPLE-10 dataset

FIGURE 4.1: Examples of the datasets used in our study. (a) Simulated SAR data from the SAMPLE-10 dataset showing target structures in controlled environments. (b) RGB satellite imagery from the EuroSAT dataset showing different land-use types. (c) SAR imagery from the MSTAR-10 dataset with differences in target type and azimuth angle.

4.2 EuroSAT Dataset

EuroSAT [16] is a satellite image dataset from Sentinel-2 with ten different land-use and land-cover classes, including cities and towns, forests, rivers and farmlands.

For consistency with SAR data, we only use the Red-Green-Blue (RGB) bands in this study. The dataset provides a shift from coherent to diffuse scattering, allowing us to test the model's cross-modality generalisation.

The dataset has high intra-class and inter-class variance, especially between visually similar land-cover classes, making it challenging to classify.

4.3 SAMPLE-10 Dataset

SAMPLE [17] is a collection of simulated SAR images and real SAR measurements for the same classes of vehicles as MSTAR. The synthetic images are produced using physics-based modelling to mimic the SAR imaging process.

However, there are differences in noise and scattering responses, resulting in a domain shift between simulated and real samples. This dataset is used to test the domain generalisation of the proposed framework from synthetic to real data.

4.4 Indian Pines Dataset

Indian Pines [18] is an agricultural scene captured by the AVIRIS sensor. This has about 200 spectral bands to support material classification.

In this study, ten classes are chosen to meet the training sample requirement (121 samples per class). Spatial patches are cropped and resized to 64×64 pixels to fit the CNN input size.

The large dimensionality of the hyperspectral data presents issues of redundancy and the Hughes phenomenon, making it a good candidate to test the proposed representation.

4.5 Dataset Summary

The summary of datasets in Table 4.1 shows the variety of sensing technology and classification problems investigated in this work. The datasets considered are SAR-based (MSTAR-10 and SAMPLE-10) data, multispectral optical data (EuroSAT) and hyperspectral data (Indian Pines), providing a broad evaluation over a range of different distributions. The datasets are composed of ten classes and a fixed number of 121 training samples per class, to simulate a low-data scenario. While there are differences in sensors and application tasks (from automatic target recognition to land-use classification), the standardised experimental protocol facilitates meaningful comparison and evaluation of model generalisation across different modalities.

Table 4.1: Summary of datasets used in this study

Dataset	**Sensor**	**Classes**	**Train/Class**	**Task**
MSTAR-10	SAR (X-band)	10	121	ATR
EuroSAT	Sentinel-2 (RGB)	10	121	LULC
SAMPLE-10	SAR (simulated)	10	121	ATR
Indian Pines	AVIRIS (HSI)	10	121	HSI Classification

4.6 Experimental Protocol

All datasets are evaluated using a standardised protocol. For all datasets, 121 samples of each class are randomly chosen for training and the rest of the samples are used for testing.

Images are resized to 64×64 and normalised to $[0, 1]$. Data augmentation and pretraining are not used, to show that the performance achieved is reflective of the inherent capacity of the proposed framework with scarce supervision.

The same preprocessing and input settings are applied to all datasets. Details of the model architecture and training is as outlined in Chapter 3.

Chapter 5

Result and Discussion

5.0.1 Cross-Dataset Generalisation

Table 5.1 shows the accuracy of the baseline CNN and our CNN-TM model on the four benchmark datasets. all four benchmarks. Our approach outperforms the baseline in all modalities, but especially for the two most the two more difficult data sets.

The largest gain is on the Indian Pines dataset, where accuracy increases from 32.05% to 84.36%. This benchmark dataset is a hyperspectral one that is difficult to 200 spectral bands and complex patterns within small patches. contiguous patches. The result shows that the structured relative spectral-spatial variations in the binary encoding accurately and that the TM infers consistent logical rules that generalise well even when training samples are scarce.

On the MSTAR dataset, the model obtains 98.39% in the cross-condition test (17° train, 15° test). This shows that the multi-scale CNN feature representation and and rule-based learning are successful in extracting edges, shadows and dominant scatterers. major reflectors, and the discrete binary interface eliminates the speckle-induced noise variations that cause conventional networks to overfit. This study shows that TMs can be used with remote sensing data, especially SAR data where interpretability and robustness are critical.

The results on EuroSAT and SAMPLE-10 also support that the model extends to SAR, optical and hyperspectral data by learning cross-modality binary features. The SAMPLE-10 accuracy of 100.00% was based on the separability of the ten electromagnetically calibrated simulated vehicle classes and not leakage; pre-visualisation

Table 5.1: Cross Dataset Results: CNN vs. CNN+TM

Dataset	CNN (%)	CNN+TM (%)
MSTAR-10	93.07	**98.39**
Indian Pines	32.05	**84.36**
SAMPLE-10	61.78	**100.00**
EuroSAT	80.74	**87.37**

Table 5.2: Comparison with Deep Learning Models on MSTAR-10 (121 training samples per class)

Model	Accuracy (%)
Proposed (CNN+TM)	**98.39**
EfficientNetV2S	89.36
EfficientNetB0	87.92
ResNet50V2	85.48
EfficientNetB2	84.66
EfficientNetB1	83.87
ResNet101V2	83.09
ResNet152V2	82.27
MobileNetV2	57.57
NASNetMobile	15.26
MobileNetV3Large	11.42
MobileNetV3Small	11.29

of the data that the sensor geometry used results in highly highly discriminative signatures from the ten classes.

5.0.2 Baselines from Deep Learning

Table 5.2 compares the proposed CNN–TM with eleven traditional deep learning networks on MSTAR, training from scratch with the same 121 samples per class. The proposed proposed framework significantly exceeds the baselines.

There are two intertwined reasons why the CNN–TM framework. First, the structured binarisation converts the continuous CNN features into binary patterns. This makes it less vulnerable to noise and overfitting to sample-specific features such as speckle, because the binarisation thresholds effectively serve as regularisers that filter out activation magnitudes while preserving rank. Second, the TM classifier uses propositional

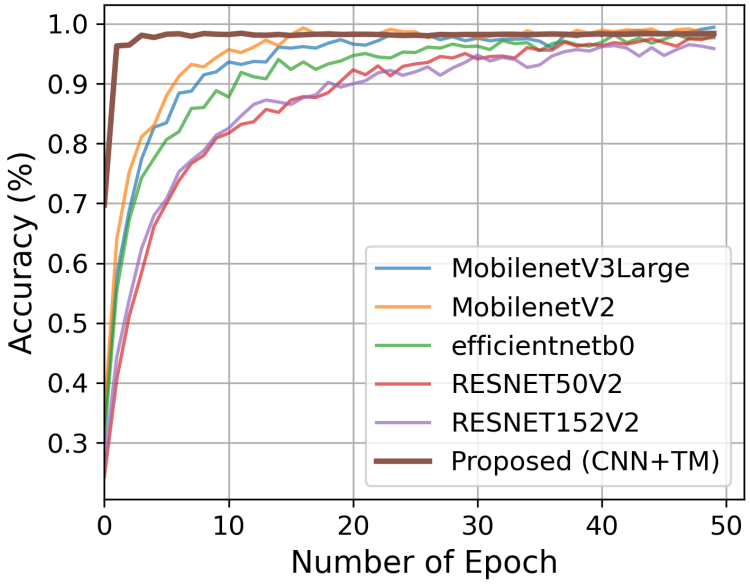


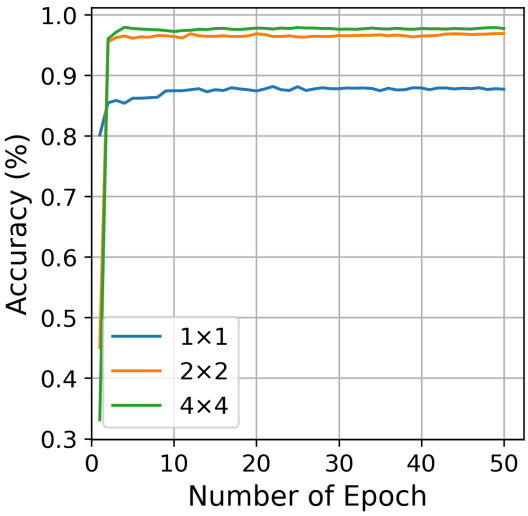
FIGURE 5.1: Convergence of the proposed CNN+TM and baseline architectures over 50 epochs of training on MSTAR-10.

logic as opposed continuous optimisation. Rather than using a large number over millions of gradient-trained parameters, it learns succinct clauses that represent patterns in the classes. This bias is in line with the low-data regime: the TM has many fewer parameters compared to its hypothesis class capacity, leading to improved generalisation.

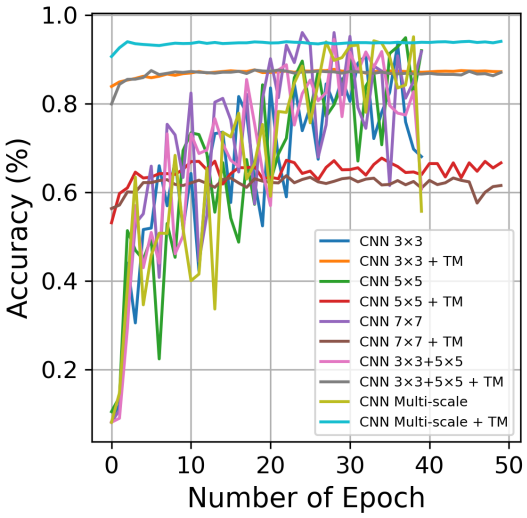
In particular, we show that even lightweight models like MobileNet and NASNet are less effective in this regime, demonstrating that compression cannot solve the problem of lack of supervision. This issue is tackled at the representation level by level rather than by scaling the architecture.

5.0.3 Convergence Behaviour

The convergence curves in Fig. 5.1 show a difference in the dynamics of deep networks and the CNN-TM pipeline. Learning curves for conventional deep networks characteristic of stochastic gradient-based optimisation with small large dropouts, and take up to 40 epochs to converge. The TM component, on the other hand, shows monotonic after 20 epochs, in line with the discrete automata update rule with no gradient oscillations. The CNN–TM pipeline thus can converge more rapidly to greater accuracy, which is useful in remote sensing applications where training needs to be fast and robust to varying acquisition conditions.



(a) Effect of spatial patch grid resolution (1×1, 2×2, 4×4) on MSTAR-10 accuracy



(b) Recognition accuracy for single-scale, two-scale and three-scale CNN front-ends with CNN and CNN+TM classifiers

FIGURE 5.2: Ablation study of the proposed system. (a) Effect of the spatial resolution of patches. (b) Impact of multi-scale CNN front-end.

5.0.4 Effect of Multi-Scale Design and Spatial Encoding

Fig. 5.2 shows two such ablation studies. The left panel compares five CNN front-end designs (single-scale 3×3, 5×5, 7×7; two-scale 3+5, 3+7; and the proposed three-scale 3+5+7) both with and without TM replacement. The multi-scale configuration performs best regardless of whether CNN or TM head, indicating that the three kernel contain complementary information: fine structural edges (3×3), intermediate texture (5×5), and broad gradients associated with shadows and specular lobes (7×7). The right panel ablates the spatial patch grid resolution (1×1, 2×2, 4×4), showing the 4×4 has the highest accuracy, which demonstrates the importance of spatial locality to classes that have similar global features but different structural layouts.

5.0.5 Interpretability and Spatial Attribution

In order to evaluate the impact of the structured binary encoding interpretability, a baseline was built where the conventional binarisation (thresholding each CNN feature map channel at its mean intensity value of each channel was applied to the extracted CNN feature maps, and then the binary vectors were used for the TM classifier. This baseline obtained a recognition rate of 98.39% on MSTAR-10, similar to our method, demonstrating that

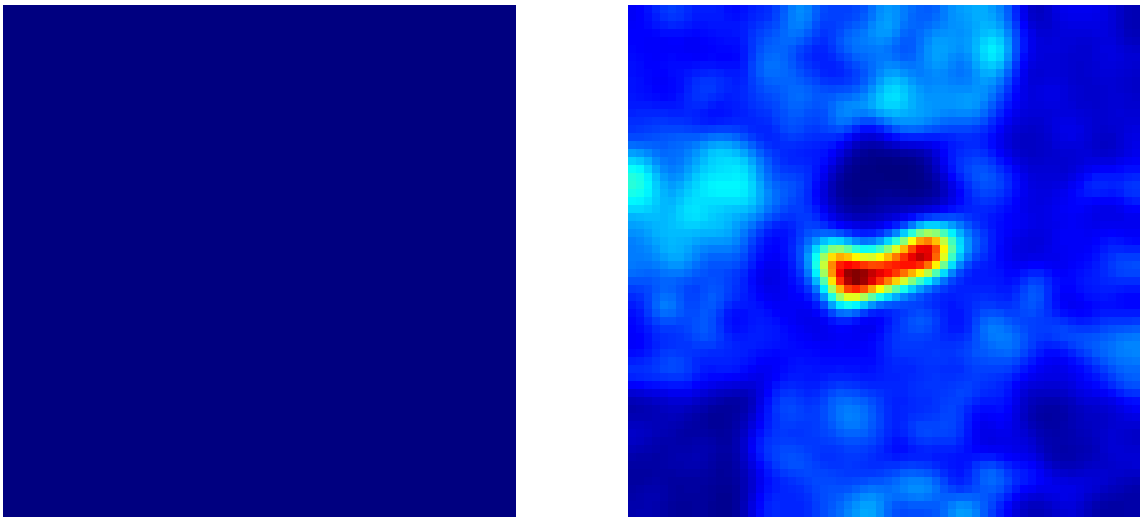


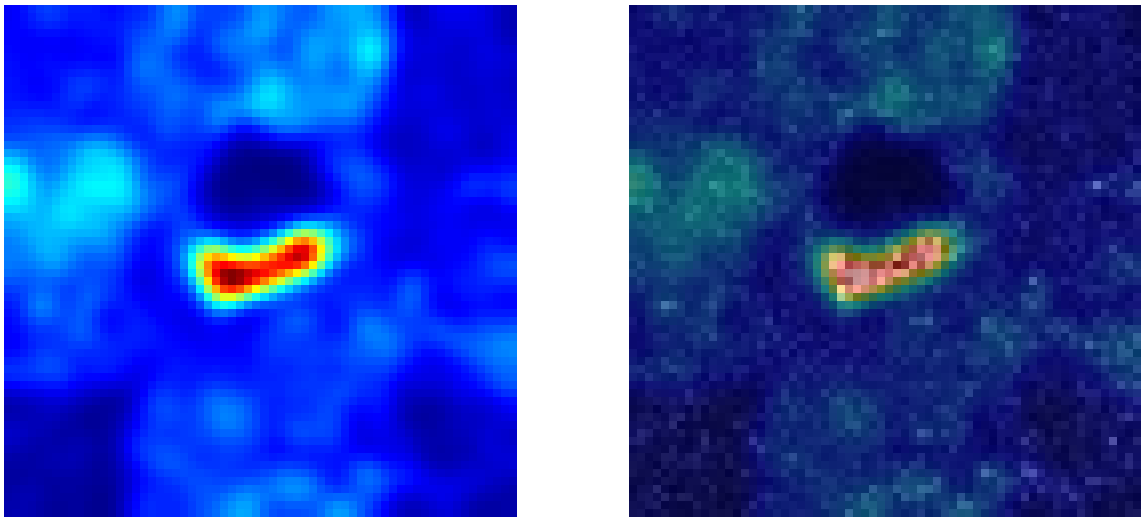
FIGURE 5.3: Attribution maps on MSTAR-10. Left: conventional binarisation (not localised). Right: new method (localised, physically meaningful regions)

.

classification performance can be achieved by simpler binarisation schemes. But as shown in standard binarisation approach is spatially diffuse and fails to conventional binarisation method is not spatially concentrated and image cannot discriminate between the target and background image. Since the global threshold ignores the spatial structure of the activity in each channel, the binary features are not informative about spatial position and the interpretation module has no way of computing spatial attributions from them.

On the other hand, the introduced structured encoding conserves local spatial statistics within each channel via normalisation and multi-threshold binarisation. This allows the interpretation module to pinpoint active features in the sources on the CNN feature map. The resulting attribution map is focused on the major scattering centres and areas of the vehicle (the hull corners and shadow boundaries which are physically consistent with the main discriminative features in X-band SAR images.

In Fig. 5.4 we show the heatmap original SAR amplitude image. The highlighted regions correspond to hull corners and shadows - two major X-band scatterers centres and shadow boundaries, respectively, which indicates that grounded. Importantly, the maps are not created from gradients or surrogates; they are derived from the features and clauses binary features and clause contributions used in classification. This feature is not something that can be provided by post-hoc that GradCAM and LIME do not offer, because they is isolated from the prediction.



(a) Normalised heatmap (b) Overlay on SAR image

FIGURE 5.4: Spatial attribution of proposed method. *Left:* normalised heatmap. *Right:* overlay on SAR image showing scatterers.

The results overall show that the proposed CNN-TM CNN-TM framework overcomes the three structural issues in Section 1: limited training data, lack of intrinsic interpretability, and sensitivity to speckle noise. By separating feature extraction from classification and using binary representations, the model combines the advantages of deep learning and rule-based systems: the CNN extracts informative spatial features from small datasets and the TM provides reliable and inherently interpretable inference.

The consistent performance gains and structurally assured attribution maps for the remote sensing data demonstrate the feasibility of the approach as a viable alternative to the deep learning only models in practical remote sensing applications.

5.0.6 Confusion Matrix and Per-Class Analysis

To gain insight into model performance, per-class and confusion accuracy, we examine confusion matrices and per-class accuracy on all datasets.

5.0.6.1 MSTAR-10 Confusion Matrix

The confusion matrix is dominated by the diagonal elements, showing the model has high classification accuracy for almost all vehicle classes. The small number of off-diagonal entries suggest that are highly non-random and infrequent.

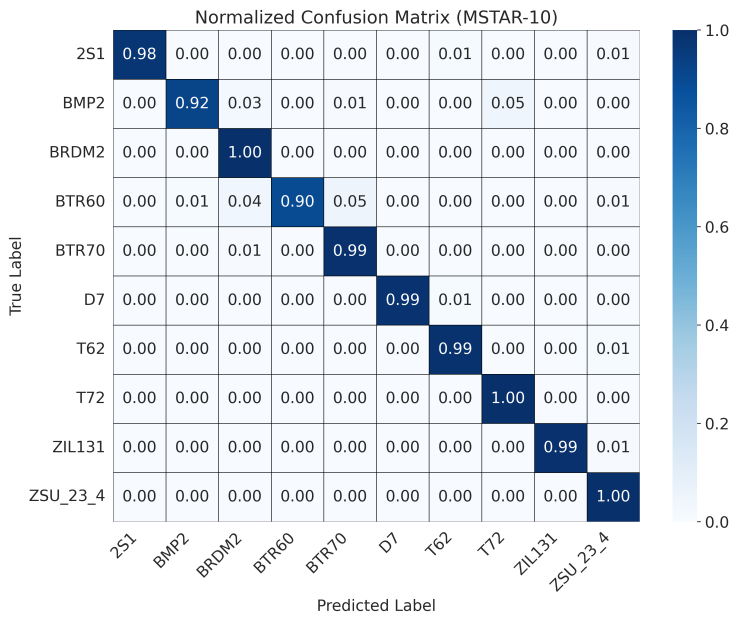


FIGURE 5.5: Normalized confusion matrix for MSTAR-10 (17°/15°)

There is some confusion between classes with similar characteristics. In particular:

- Armored vehicles with similar-shaped turrets exhibit occasional overlap.
- Minor differences in the viewing angle distort the scattering patterns, resulting in small errors.

These findings show that the errors are due to intrinsic similarity, not model instability.

5.0.6.2 Accuracy Per Class

The class-level analysis shows that the performance is not distributed across classes. but rather is highly sensitive to inter-class similarity and data characteristics.

Indian Pines: Results vary greatly, with spectral similarities. vegetation classes. Improvements are in class confusions hence the big jump in accuracy.

EuroSAT: Land use types such as *Forest* and *Industrial* are accuracy, and *Highway* and *PermanentCrop* are less accuracy because of visual similarities.

MSTAR-10: Most classes have an accuracy over 95%, showing good structural discrimination capability.

Table 5.3: Per-class accuracy for all datasets

Class	Indian Pines	EuroSAT	MSTAR-10	SAMPLE-10
Class 1	0.91	0.67	0.95	0.88
Class 2	0.72	0.88	0.98	0.91
Class 3	0.59	0.67	0.96	0.60
Class 4	0.58	0.39	0.94	0.58
Class 5	0.88	0.86	0.99	0.89
Class 6	0.94	0.75	0.97	0.93
Class 7	0.82	0.53	0.98	0.82
Class 8	0.94	0.77	0.99	0.94
Class 9	0.88	0.67	0.97	0.88
Class 10	0.93	0.81	0.99	0.93

SAMPLE-10: Almost perfect accuracy due to high separability of simulated electromagnetic signatures.

Overall, errors are concentrated in semantically or structurally similar classes, rather than being randomly distributed.

5.0.7 Comparison with Direct Tsetlin Machine Variants

To evaluate whether the CNN feature extraction stage is essential, we compare the proposed CNN–TM framework against a direct Tsetlin Machine-based approach, specifically the Convolutional Tsetlin Machine (ConvTM), which operates directly on image inputs without requiring a learned feature extractor.

The ConvTM configuration used in this study follows the setup reported in [19], a well-validated implementation applied to structured 5G signal data. The model parameters are fixed as:

- `number_of_clauses` $= 200$
- `T` $= 477$
- `s` $= 2.081$
- `patch_dim` $= (10 \times 10)$
- `max_included_literals` $= 22$

The correctness of the implementation is verified by reproducing results on the original domain, where the ConvTM achieves an average accuracy of $95.45 \pm 2.03\%$ on 5G signal data, consistent with the reference study. Additional validation from [20] reports accuracy in the range of 97.4%–97.8% for hardware-based TM implementations, confirming that the architecture is capable of strong performance when applied to structured inputs.

The same ConvTM configuration is then evaluated on the MSTAR-10 dataset under the identical experimental protocol used throughout this work, namely 121 training samples per class. Due to the stochastic nature of Tsetlin Machine training, results are averaged over ten independent runs.

TABLE 5.4: Comparison of Tsetlin Machine-based models across domains. Paper-reported results are shown alongside reproduced implementations and the proposed CNN–TM model.

Source	**Dataset**	**Model**	**Avg Acc (%)**	**Max (%)**	**Std Dev**
[19]	5G SSB	ConvTM	95.45	—	2.03
[20]	FPGA benchmark	TM	97.6	97.8	—
Ours (Reproduced)	FPGA benchmark	TM	48.61	50.93	1.19
Ours (Reproduced)	MSTAR-10	ConvTM	20.80	—	4.62
Ours (Proposed)	MSTAR-10	CNN–TM	**98.39**	—	—

A detailed breakdown of the ConvTM performance across runs is provided in Table 5.5.

TABLE 5.5: Per-run ConvTM accuracy on MSTAR-10 under the 121-sample-per-class protocol

Run	**Accuracy (%)**
Run 1	21.53
Run 2	14.19
Run 3	24.70
Run 4	25.36
Run 5	21.73
Run 6	18.27
Run 7	18.14
Run 8	12.54
Run 9	26.64
Run 10	24.87
Mean	20.80
Std Dev	4.62

Nevertheless, one should notice the dramatic decrease in performance of the ConvTM approach when dealing with SAR images instead of signal data. While ConvTM works quite successfully in its native setting, it achieves an average accuracy of $20.80 \pm 4.62\%$

in classification on MSTAR-10 dataset, outperforming random classification with only a marginally better score.

The low standard deviation of the accuracy shows that this trend is predictable, indicating the influence of some inherent properties of the model in question. Namely, such an accuracy decrease implies the inability to identify the features of data in the context of limited supervised learning in the SAR images domain.

Such a property of the model can be attributed to the characteristics of the input data. In the case of working with the 5G signal data, the model deals with patterns in the data and thus manages to extract features, even after their binarization. However, the SAR images involve speckle noise alongside useful signals, which appears in the intensity domain.

Nonetheless, the application of binarisation on SAR images directly results in noise being represented along with the structural data leading to a non-semantic binary form, whereby noise prevails. In such circumstances, the learning algorithm does not capture any semantic structure in the data but only learns noise; therefore, it provides very poor results.

A solution to this problem is provided by the front-end network used in the current model which performs hierarchical feature extraction prior to binarisation. Filtering performed in the convolution layer eliminates the noise from the input data, allowing identifying stable elements in the image like edges, shadows, scattering, etc. Hence, such intermediate data becomes more suitable for clause training rather than binarisation.

As a result of the process performed by the convolutional front-end network, learned logical rules turn out to be consistent and correspond to concrete elements in the SAR image. Therefore, a significant increase in the quality of the algorithm is provided since the accuracy achieved by the proposed method equals to 98.39%, which shows 77 percentage points improvement compared to the ConvTM approach. These results reinforce two key conclusions:

- The Tsetlin Machine is highly effective when provided with structured and meaningful binary representations.
- Direct application of TM variants to raw SAR data is not sufficient; a feature extraction stage is essential.

Overall, this comparison validates both the correctness of the implementation and the necessity of the proposed hybrid CNN–TM architecture for remote sensing applications.

5.0.8 Case Study: Correct and Misclassified Predictions

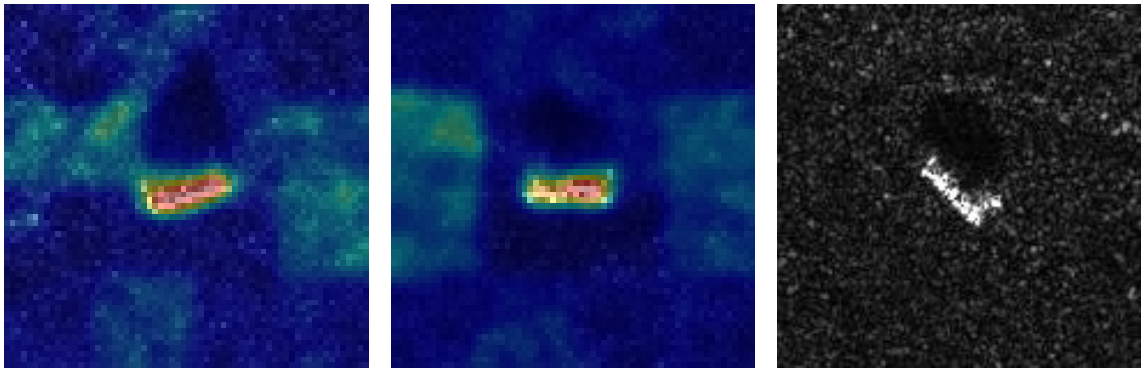
A qualitative study will be performed on sample cases from each dataset for a more detailed analysis on how the CNN-TM works. As demonstrated in Fig. 5.6, each row consists of one image that the model classified correctly, one image that the model misclassified, and the class of the latter image.

Starting with the correctly classified case, it can be seen that the model has learned well enough to discriminate between the spatial patterns and scatters in the target. When it comes to datasets such as MSTAR-10 and SAMPLE-10 where the classification is based on SAR imagery, the model is capable of learning the structural features to differentiate between the main scattering centers and geometric configuration. In addition, when dealing with datasets such as EuroSAT and Indian Pines, the model is able to classify targets based on texture and spectral data.

The misclassification problem can be attributed to having high similarity in images of the same category and/or images of different categories that have similar appearances. In the case of SAR imagery, misclassification happens due to similar geometric configuration and observation angles, making it hard to differentiate the scattering pattern. This applies especially to EuroSAT and Indian Pines datasets on optical imagery.

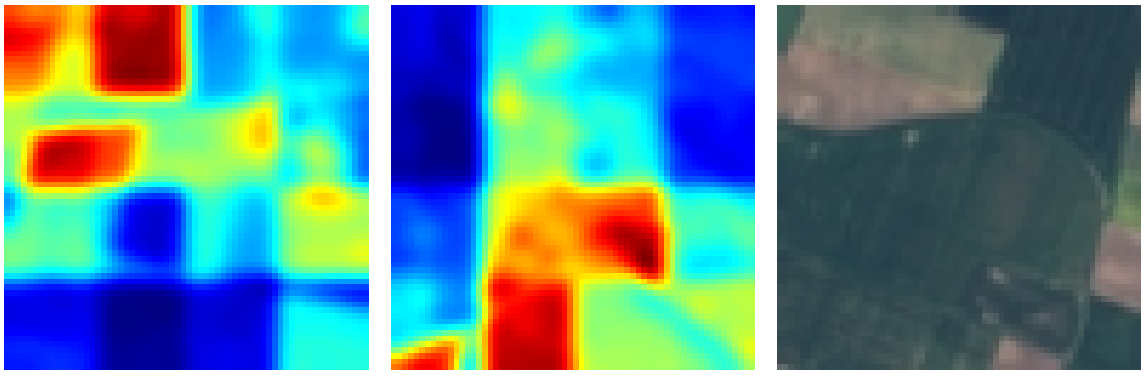
It can be deduced from the above results that the new model works effectively by creating good spatial representations. Its effectiveness varies, depending on slight structural differences and learning conditions. The important thing about the new model is to ensure the existence of both right and wrong predictions for the same set of data because this proves that the errors are consistent with the data used.

MSTAR-10



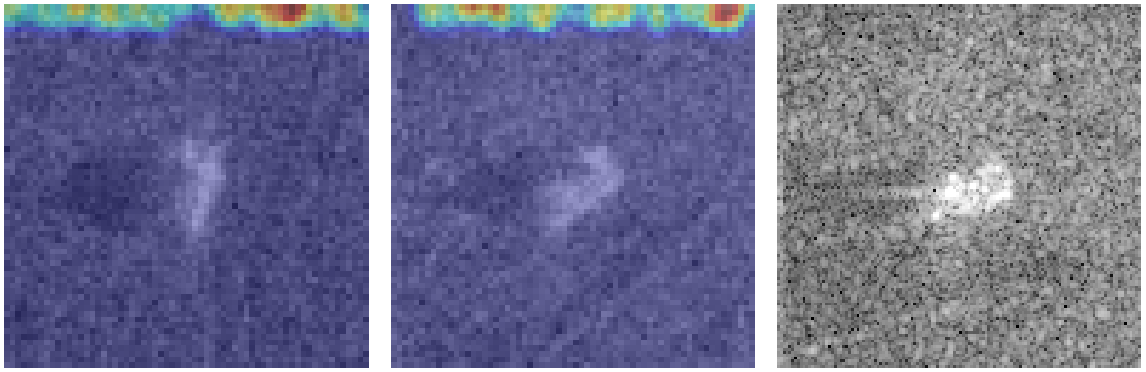
(a) Correct - 2S1 (b) Incorrect - ZIL131 (c) Sample - 2S1

EuroSAT



(d) Correct - Annual Crop (e) Incorrect - Pasture (f) Sample - Annual Crop

SAMPLE-10



(g) Correct - ZSU23 (h) Incorrect - M35 (i) Sample - ZSU23

FIGURE 5.6: Comparison of correct and incorrect predictions across datasets. Each row shows (left) correct prediction, (middle) incorrect prediction, and (right) a representative sample image from the dataset.

Chapter 6

Conclusion

6.1 Conclusion

In this paper, we propose a new architecture called CNN-Tsetlin Machine (CNN-TM), which aims to address three major issues in the domain of satellite image classification. These are a limited number of labelled training samples, vulnerability to sensor noise, and interpretability.

By splitting the two steps, CNN-TM leverages a multiscale convolutional neural network (CNN) to learn high-level representations and relies on the Tsetlin machine to provide generalisation and interpretability via propositional clause learning. Additionally, the structured binary encoding module helps link these two components by transforming the multi-scale CNN's output to channel-normalised binary representations while retaining its relative strength, texture, and maximum values.

The most crucial advantage of our design is the ability to generate the attribution map based on the clause set used to predict, thus ensuring consistency between the structures of both procedures. Therefore, there is no need to employ other approaches to interpret the results.

We evaluate our model against several state-of-the-art deep learning methods using four sensing modalities under a tight limit of 121 samples per category. The model achieves an accuracy of 98.39% on MSTAR-10, surpassing all other baseline methods such as EfficientNetV2S (89.36%) and ResNet50V2 (85.48%).

6.2 Future Work

Some possible extensions of the proposed technique are listed below.

- **Polarimetric and Complex-Valued Extensions:** A possibility to apply the proposed technique with respect to complex-valued SAR imagery and polarimetric SAR imagery is also relevant in order to achieve a better scattering process representation.
- **Online Learning:** It would be reasonable to develop an opportunity for the Tsetlin Machine to work in non-stationary conditions by applying an online learning technique.
- **Generalization Over Multiple Sensing Modalities:** Joint training over multiple sensing modalities might be helpful in order to achieve better results in heterogenous conditions.
- **Edge-Based Approach:** As the proposed technique has the lightness and interpretability properties, it could also be considered for application on satellites.

Chapter 7

Publications

This chapter summarises the research contributions resulting from the work presented in this thesis, along with related publications developed during the course of the programme.

7.1 Major Publication

P. Thangaraj, S. Chakraborty

"A Novel Convolutional Tsetlin Architecture for Classification of Satellite Images Under Limited Supervision"

Submitted to the **34th ACM SIGSPATIAL Conference**, International Conference on Advances in Geographic Information Systems, Riverside, CA, USA, November 3–6, 2026.

Status: Under Review **Tier:** Core A level

The current work represents the key novelty of the thesis and suggests an interpretable approach to satellite images classification based on a novel combination of Convolutional Neural Networks and Tsetlin Machines operating in the supervised setting. The suggested method ensures a high classification performance along with natural interpretability of results.

7.2 Other Publications

7.2.1 Journal Publications

- S. M. Akella, **P. Thangaraj**, S. R. Yeduri, and L. R. Cenkeramaddi, *"5GDC — Estimating Drone Count using 5G CSI Measurements and Multi-Channel CNN,"* Journal of Ambient Intelligence and Humanized Computing, 2026. **Status: Published [21]**

7.2.2 Conference Publications

- **P. Thangaraj**, S. R. Yeduri, O. C. Granmo, and L. R. Cenkeramaddi, *"Interpretable and Intelligent DDoS Attack Detection Method using Tsetlin Machine,"* Proceedings of the International Symposium on the Tsetlin Machine (ISTM), IEEE, 2025. **Status: Published [22]**
- **P. Thangaraj**, B. Pardhasaradhi, P. Srihari, and L. R. Cenkeramaddi, *"A Lightweight Dual-Residual CNN for Accurate Human Motion Classification Using Multi-Sensor IMU Data,"* IEEE International Symposium on Smart Electronic Systems (iSES), 2025. **Status: Accepted**

7.2.3 Under Review

- **P. Thangaraj** et al., *"Clustering-based Significance-Aware Automated Data Binarization Framework for Tsetlin Machines,"* IEEE Transactions on Artificial Intelligence. **Status: Under Review**
- **P. Thangaraj**, S. P. Narayanan, and L. R. Cenkeramaddi, *"Thermal Imaging-based Drone Detection using Lightweight YOLO Architectures,"* IEEE Sensors. **Status: Under Review**

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